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HQML-NLP: A hybrid quantum machine learning framework for scholarly AI-text detection

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HIGHLIGHTS

- Proposes HQML-NLP, a hybrid quantum–classical NLP framework.
- Integrates SBERT embeddings with a 6-qubit parameterized quantum circuit.
- Achieves AUC > 0.96 on AI-GA, HWAI, and large-scale HAGT-1M datasets.
- Uses temperature scaling to reduce calibration error by up to 60 %.
- Matches prior accuracy with over 2000 × fewer parameters than transformers.

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ABSTRACT

The swift growth of the gargantuan language models has made it even harder to tell apart writings done by humans and AI especially in academia which led to the need for the setup of trustworthy detection frameworks that will satisfactorily balance accuracy, interpretability, and efficiency. In this paper, we walk you through the HQML-NLP, a detection system using a hybrid quantum-classical machine learning framework for the detection of AI-generated academic content. The system onboard the merging of Sentence-BERT semantic embeddings with quantum feature encoding that is supported by a 6-qubit, two-layer parameterized quantum circuit, thus resulting in a 390-dimensional hybrid representation which is classified through a lightweight multilayer perceptron. The framework has been tested on three benchmark datasets AI-GA, HWAI, and HAGT-1M to check its scalability and generalization to different academic writing situations. The results of the experiments show the framework has consistent and good discriminative capability achieving AUROC scores of more than 0.96 on all datasets and excellent performance (AUROC = 1.000, ACC = 99.98 %) on the large-scale data. Moreover, probability calibration by means of temperature scaling raises the trustworthiness of predicted confidence scores, leading to a 60 % reduction in the Expected Calibration Error (ECE) and without affecting the performance of the discrimination. When compared against the transformer-based and ensemble-learning detectors, HQML-NLP comes out with an equivalent and competitive detection accuracy and calibration quality yet demands more than 2000× less trainable parameters. These findings imply that hybrid quantum-classical representations act as an effective and compact alternative for the detection of AI-text in scholarly journals.

1. Introduction

OpenAI's GPT-4, Anthropic's Claude, and Google's Gemini are powerful large language models, which have really influenced the landscape of generating texts to the extent that now even human-like reasoning plus the highest quality of linguistic fluency in both academic and professional arenas can be imitated. However, also being the case, these technological advancements brought about problems of a considerable magnitude regarding the integrity of academic practices, verification of authorship, and authenticity of content [1–3]. AI-assisted production of scholarly texts has become so good that it can even imitate the use of specialized jargon, the following of specific rhetorical

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1568-4946/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

patterns, and the employing of domain-specific arguments, thus creating confusion over the line between real and artificial authorship [4]. This scenario indeed brings forth the necessity of very strong, well-explained, and highly scalable detection and discrimination systems which would be able to tell the difference between AI-generated and human-written academic texts [5,6].

Authenticity models of traditional text largely depended on statistical stylometry, lexical heuristics, and perplexity-based metrics that were based on probabilistic language models [7,8]. Those methods, however, encounter difficulties with transformer-generated contents as they possess both high contextual coherence and low perplexity. In a bid to address these issues, the scholars applied ML and DL techniques going from logistic regression and random forests to recurrent and convolutional networks in order to capture [9–11], sign of linguistic richness. Still, as the generative models become more and more advanced, even deep networks have been unable to keep discriminative stability without huge computing power [12,13].

The advent of transformer-based architectures, especially BERT, RoBERTa, and Sentence-BERT (SBERT), has made a giant leap for feature extraction in natural language processing (NLP) [14]. The new embeddings created are semantically context-aware thus making it easier to detect even the most subtle stylistic differences between genuine and faked texts [15,16]. Following this trend, Hazim and Ata [17] put forward the revelation of Textual Authenticity in the AI Era, where they compared the two embedding methods SBERT and RoBERTa through logistic regression and feedforward neural networks for the text detection of AI. Their research has surpassed the 99.9 % accuracy mark on academic datasets that had been balanced, which demonstrates the effectiveness of transformer embeddings but at the same time reveals some important drawbacks like very large parameter counts, the need for inefficient training, and lack of calibration reliability. To tackle the interpretability of the model and the computational burden, Hazim et al. [18] proposed the use of an NLP Framework based on Ensemble Learning, which secured almost the same level of accuracy ($\approx$ 98.7 %) but with a much lower computational load owing to the application of classical ensemble methods. Although the two frameworks were a significant step forward in the development of the authentication of text, they still relied completely on classical methods, hence, the possibility of using quantum feature spaces to represent the non-linear linguistic entanglements was not explored.

The newest studies in Quantum Machine Learning (QML) provide a cutting-edge viewpoint for text analysis based on the use of quantum superposition and entanglement to represent the correlations of very high dimensions which classical computation cannot [19–21]. Hybrid quantum-classical systems bring together parameterized quantum circuits (PQCs) with neural components, thus increasing the potential for interpretation while lowering significantly the costs for parameterization and electricity [22,23]. Although QML showed its potential in a number of applications, such as image recognition and signal processing, textual authenticity detection is still one of the least explored areas, mainly due to the fact that academic discourse is characterized by stringent linguistic demands and the need for interpretability [24].

In support of this statement, a systematic review of the primary academic databases (Scopus, IEEE Xplore, SpringerLink, and ScienceDirect) for the years 2019–2025 found just one study that was directly related to the detection of AI-generated content using hybrid or quantum-enhanced methods. The lack of prior work in existence confirms the novelty and the timeliness of the current research and indicates the lack of extensive empirical frameworks in this new area. Authors believe that no exploration has been done to propose and verify a hybrid quantum–classical NLP architecture for the specific task of detecting AI-generated academic text. Consequently, this work identifies HQML-NLP as a first-of-its-kind contribution connecting classical semantic embeddings and quantum feature spaces to push the envelope of reliable AI-assisted authorship verification even further.

Motivated by the gaps identified, the present study launches HQML-NLP, a Hybrid Quantum–Classical Machine Learning system which is intended for the detection of AI-text in the scientific domain. The set-up combines SBERT semantic embeddings (384-dimensional) with a six-qubit, two-layer PQC to generate a 390-dimensional hybrid feature space that includes both semantic depth and quantum-entangled structural cues. The model is trained and evaluated on three different academic datasets AI-GA, HWAI, and HAGT-1M which cover a wide range of domains and scales, thus allowing for a rigorous evaluation of scalability, calibration reliability, and generalization. Fig. 1 illustrates the conceptual workflow and application context of the proposed framework, while the detailed computational and algorithmic components are presented in Fig. 2.

This paper presents the following key contributions:

1. We put forward HQML-NLP, a hybrid quantum–classical natural language processing framework that not only combines the Sentence-BERT semantic embeddings but also utilizes parameterized quantum circuits for the purpose of identifying the AI-generated scholarly text.
2. We set up a thorough evaluation involving multiple datasets, namely, AI-GA, HWAI, and HAGT-1M, thus creating empirical benchmarks for the hybrid quantum–classical techniques applied in the academic context of author verification.
3. We employ probability calibration through temperature scaling, and we evaluate the prediction reliability using Expected Calibration Error (ECE) in a systematic manner, thus bringing about the comprehensibility and reliability of detection results.
4. We present a complete comparative study against the previous transformer-based and ensemble-learning frameworks, which shows that HQML-NLP acquires the same discriminative performance as a new method with just over 2000 $\times$ fewer parameters and better calibration stability.

This paper consists of several sections. Second section reviews the literature on AI-generated text detection, transformer and hybrid approaches and quantum NLP. The third section presents the HQML-NLP methods and architectural design. The fourth section reports experimental results and comparative evaluations across datasets. The fifth and final section concludes with a discussion of the implications, limitations, and future research directions.

2. Related work

The development of AI-generated text detection research has gone through various methodological stages, which mirrored the growing complexity of large language models (LLMs). At first, the methods were based on linguistic characteristics made by hand and statistical stylometry, and then they were complemented with transformer-based contextual representations and, recently, hybrid and quantum-inspired learning frameworks. By pinpointing the most important trends, restrictions, and unresolved issues, this part brings together the past studies that drive the need for the HQML-NLP framework.

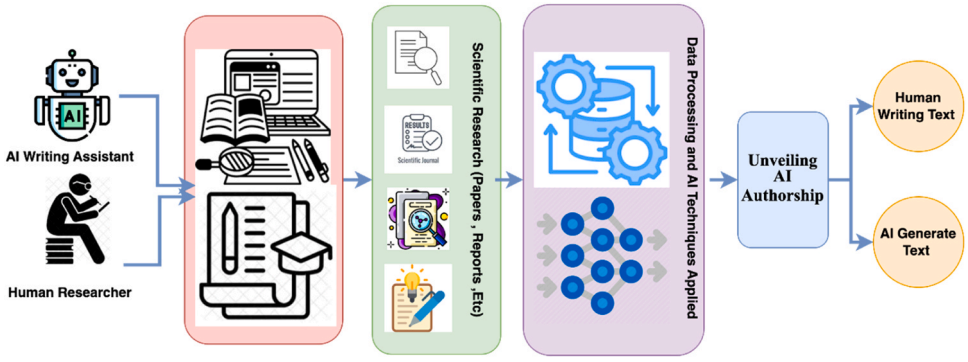


Fig. 1. High-level conceptual overview of the HQML-NLP workflow.

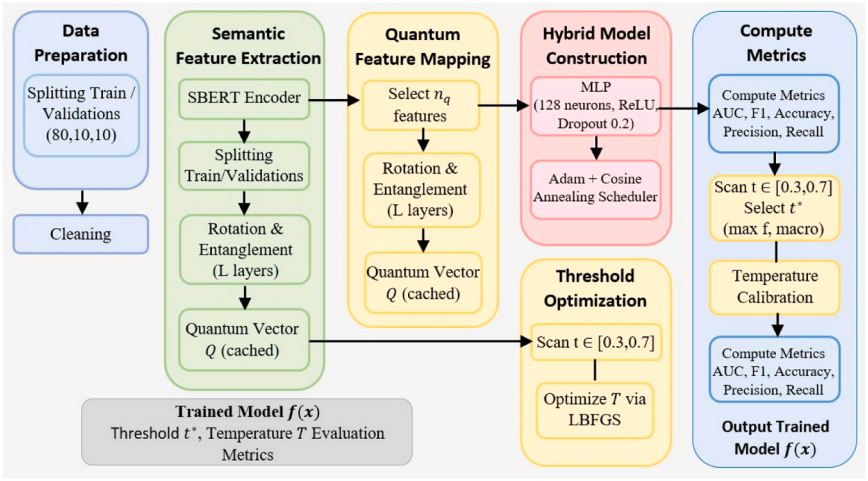


Fig. 2. Proposed HQML-NLP framework.

2.1. Evolution of AI-generated text and detection challenges

The evolution of text generation has progressed rapidly it has moved from the very simple rule-based and statistical n-gram models to the very complex deep neural networks and large transformers. The groundbreaking models like GPT-2 and GPT-3 have shown a unprecedented levels of fluency and coherence and coherence in the context (Radford et al. [25]; Brown et al. [26]), and this was the start of the development of even more powerful LLMs. Their successors GPT-4 (OpenAI, [2]), Claude (Anthropic, 2024), and Gemini (Google DeepMind, 2023) have not only inherited these abilities but also extended them to the areas of academic, scientific, and professional writing, allowing the models to produce specialized arguments, organized abstracts, and stories similar to research ones in their respective fields.

The transformation of the paradigm has led to the convergence of real and fake scientific authorship. Research by Zellers et al. [9] and Gao et al. [8] showed that even expert reviewers can be led astray by AI-generated research abstracts, thus, the importance of having strong and understandable detection systems was highlighted. With AI producing more and more scholarly content that is not only large in quantity but also sophisticated in quality, the task of telling apart human-written and machine-generated academic text has become the primary issue faced by computational linguistics, digital forensics, and research integrity analysis.

2.2. Classical, transformer-based, and hybrid detection approaches

Early AI-text detection methods relied primarily on handcrafted linguistic features, statistical stylometry, and frequency-based representations such as TF-IDF and n-grams. Classical classifiers, including Naïve Bayes, Logistic Regression, Support Vector Machines, Decision Trees, and Random Forests, were applied to identify stylistic and lexical differences between human- and machine-generated text (Kowalczyk et al. [27] and Singh et al. [28]). The methods used in this study produced decent results for lengthy or heavily patterned texts, but their dependence on superficial characteristics resulted in a lack of strength when it came to the various fields and when the content generated by transformers had low perplexity.

The application of transformer-based encoders to AI-text detection has been a radical development as it made possible the acquisition of contextualized semantic representations through self-attention mechanisms. BERT, RoBERTa, and Sentence-BERT (SBERT) are the leading models that captured not only the sentence-level semantics, discourse structure, and stylistic regularities, but also the superior performance (Ravindhran [15]; Lak A et al. [14]; Francisca Adoma A et al. [29]). Grover [9] was an example of the technology that could do both: making synthetic text look good and detecting it; at the same time, the studies that followed did not fail to report improvements on classical baselines in the area of fake-news and scientific-content detection consistently [15,16,30].

Following the line of thinking, Hazim and Ata [17] besides support from SBERT and RoBERTa embeddings combined them with Logistic Regression and Feedforward Neural Networks, getting accuracy close to perfection on extensive academic datasets. Nonetheless, the research also pointed out some drawbacks of transformer-based detectors, which include high computational cost, over-parameterization, drifting calibration, and being hard to interpret.

To overcome these problems, the fusion of classical machine learning, deep networks, and ensemble strategies has been suggested as a solution. The ensemble-based framework by Hazim and Ata [18] ("Bridging the Gap") proved that the combination of carefully designed classical features and multiple classifiers can result in the same level of accuracy as with much simpler methods. Further, the other hybrid architectures with CNNs, RNNs, and optimization algorithms have even more enhanced the cross-domain robustness [31,32]. Although explainability techniques such as SHAP and LIME enhanced transparency [33,34], they introduced additional computational overhead. Overall, existing hybrid systems remain fully classical and struggle to efficiently model complex, non-linear semantic interactions inherent in scholarly text.

2.3. Emerging quantum-enhanced NLP approaches

Quantum Machine Learning (QML) has attracted a lot of interest lately and is considered among the most promising approaches for the linguistics of high-dimensions modeling via techniques like superposition and entanglement (Schuld & Killoran, [35]).

The field of computer vision (Senokosov et al. [36]) and signal-processing (Sagingalieva et al. [37]), saw major breakthroughs due to QML but its use-in-NLP is still more of an academic curiosity which only has a handful of studies looking into it [38–40].

Another development by Sumithra et al. [41] presented a DF-QCNN (Deep Fast Quantum Convolutional Neural Network), a quantum-inspired architecture, which was trained by the Crayfish Optimization Algorithm. Incorporating quantum operations into AI-generated text detection is a novel approach but it still limits itself to small datasets and does not use contextual embedding or quantum-classical hybrid feature mapping.

The current state of quantum-enhanced NLP remains largely exploratory. Existing studies primarily focus on small-scale prototypes or quantum-inspired architectures and do not incorporate contextual semantic embeddings or principled quantum–classical feature integration. To the best of our knowledge, no prior work has systematically combined transformer-based semantic representations with a learnable parameterized quantum circuit for the detection of AI-generated scholarly text. In addition, there are still some constant difficulties such as over-parameterization, high computational cost, poor probabilistic calibration, and limited interpretability among the classical, transformer-based, and hybrid detection frameworks. Though the previous works of the authors created powerful ensemble-based and transformer-based baselines [17,18], these methods work only in classical feature spaces. The proposed HQML-NLP framework fills this gap by presenting a reliable quantum-classical combination that projects semantic SBERT embeddings into a small quantum feature space, thus allowing enhanced calibration, interpretability, and parameter efficiency over the state-of-the-art detection methods.

3. Proposed framework design

We suggest HQML-NLP, which is a combination of a hybrid quantum and classical natural language processing framework that utilizes the contextual semantic embeddings along with a light, trainable quantum feature map for the detection of AI-generated scholarly text. The framework processes the input text through the use of Sentence-BERT (SBERT; all-MiniLM-L6-v2) that gives 384-dimensional semantic representations to the text. These representations are able to convey not only the meaning of the sentences but also the structure of the discourse. The preprocessing involves the normalization of these embeddings followed by a partial projection onto a parameterized quantum circuit (PQC) in PennyLane, which results in a low-dimensional quantum descriptor Q consisting of 4–6 expectation values.

The original SBERT embeddings are concatenated with the quantum features to create a single hybrid representation $[Q\|Z]$, which is then passed to a lightweight multilayer perceptron (MLP) classifier with dropout regularization. To enhance robustness and probabilistic reliability, the framework applies validation-based threshold tuning and post-hoc probability calibration through temperature scaling. The performance of the model is assessed through the use of stratified 80/10/10 train-validation-test splits and is reported in terms of Accuracy, Precision, Recall, F1-macro, AUROC, Expected Calibration Error (ECE), and confusion matrices.

Fig. 2 illustrates the complete HQML-NLP architecture, while

Algorithm 1 details the end-to-end training and evaluation procedure.

Algorithm 1. Proposed Strategy of HQML-NLP

Input: Dataset D (text, label); number of qubits n_q; number of PQC layers L; batch size B; epochs E
Output: Trained hybrid model $f(x)$; optimal threshold t^*; temperature T; evaluation metrics

Begin

Step 1: Data Preparation
- Clean text (remove duplicates, missing entries).
- Split dataset into 80% train, 10% validation, 10% test (stratified).

Step 2: Semantic Feature Extraction
- Generate 384-dimensional SBERT embeddings for each text sample.
- Standardize all features using training-set mean and variance.

Step 3: Quantum Feature Mapping
- Select the first n_q standardized features per sample.
- Encode them into n_q qubits using rotation gates.
- Apply L layers of parameterized quantum rotations followed by ring-CZ entanglement.
- Compute Pauli-Z expectation values to form the quantum feature vector Q.
- Cache Q for reuse during validation and testing.

Step 4: Hybrid Feature Fusion
- Concatenate SBERT embeddings (Z) and quantum features (Q) to obtain $[Q\|Z]$.

Step 5: Model Initialization
- Define MLP with 128 hidden neurons, ReLU activation, dropout 0.2, and sigmoid output.
- Initialize optimizer (Adam) and learning-rate scheduler (Cosine Annealing).

Step 6: Model Training
- For each epoch $e \in [1, E]$:
  a. Train mini-batches of size B using binary cross-entropy loss.
  b. Evaluate validation AUROC after each epoch.
  c. Save model if AUROC improves; otherwise increase patience counter.
  d. Stop early if no improvement is observed for six consecutive epochs.

Step 7: Threshold Selection
- Evaluate thresholds $t \in [0.3, 0.7]$ on validation data.
- Select t^* yielding highest F1-macro.

Step 8: Probability Calibration
- Convert validation predictions to logits.
- Optimize temperature T via LBFGS to minimize validation logits.
- Apply T to test logits for calibrated probabilities.

Step 9: Testing and Evaluation
- Compute test metrics using calibrated and uncalibrated outputs at t^*.
- Generate ROC curves, reliability diagrams, and confusion-matrix.
- Save train model and evaluation results.

End

HQML-NLP's computational complexity is mainly determined by three factors: semantic embedding extraction, quantum feature mapping, and classification. Let N stand for the number of samples and d= 384 the dimension of SBERT embedding. Semantic embedding extraction grows linear as $Q(N \bullet d)$.

Quantum feature creation is carried out on a fixed number of qubits n_q and circuit layers L which together result in $Q(N \bullet n_q \bullet L)$, where n_q and L are small constants in all of the experiments. The MLP classifier releases the training cost of $Q(E \bullet N \bullet h)$, with E as the number of epochs and h the number of hidden neurons. Memory consumption is mostly caused by embedding storage and cached quantum features which are thus linearly dependent on N.

To sum up, HQML-NLP shows linear time and space complexity with a minor parameter footprint (≈50k parameters), which makes it a computationally efficient and easily scalable to large academic corpora.

3.1. Datasets description

The present research employs three complementary datasets that have been specifically designed to represent different types of academic writing and AI-generated scholarly content in various fields. These datasets, when used together, allow for an extensive assessment of the scalability, robustness, and generalization of AI-text detection.

3.1.1. AI-GA dataset (academic abstracts)

The AI-GA dataset contains 28,662 abstracts (14,331 human-written and 14,331 AI-generated) derived from CORD-19 articles and their GPT-generated counterparts [42]. A (title, abstract, label) tuple constitutes every entry, wherein label = 0 signifies a human-authored text and label = 1 an AI-generated one. Preprocessing deletes markup, special tokens, and generic stop-words. This corpus offers a concentrated benchmark for studying scholarly abstract-style writing.

3.1.2. HWAI dataset (human–AI academic writing)

The HWAI dataset consists of 44,162 publications that are 22,081 each of human (HGT) and machine (MGT) generated texts [28]. The dataset records are in the form of (Text, Class), where Class ∈ {0,1}. The HWAI dataset not only covers encyclopedic and academic subjects but at the same time challenges the model generalization across larger scholarly areas.

3.1.3. HAGT-1M dataset (human–AI generated text)

The HAGT-1M dataset consists of a million balanced human–AI text pairs which are available for large-scale evaluation. The selection process for this study began with 500,000 samples, out of which 446,646 instances were saved for the HQML experiments after going through preprocessing and filtering stages (approximately 357k training, 44k validation, and 44k testing samples) [31]. Each instance has four components: (id, human_text, ai_text, instructions) and these components are combined into (text, label) pairs. The dataset is a means for large-scale analysis and simultaneously stress-tests the scalability of hybrid quantum-classical learning.

In a way, the datasets together form a range of writing domains that are overlapping each other: AI-GA is for academic abstracts, HWAI is for scholarly exposition and HAGT-1M is for large-scale generalization. The conjoined use of these datasets has given the opportunity to test HQML-NLP in varied linguistic and disciplinary conditions. conditions.

3.2. Pre-processing and data splits

The input text is subjected to a minimum normalization process which maintains the stylistic and semantic features that are necessary for the purpose of authorship detection. Preprocessing actions consist of making the text lowercase when appropriate, discarding redundant entries, and eliminating samples that are null or corrupted. The application of aggressive linguistic normalization (e.g., stemming or lemmatization) is not considered as such methods might suppress the stylistic cues that differentiate the text written by humans from the one generated by AI. The training (80 %), validation (10 %), and test (10 %) subsets are created from all datasets using stratified splits, which provide an assurance of equal class distributions across the splits. A fixed random seed (SEED = 42) is strictly maintained to achieve reproducibility through the entire process.

Stratification in the AI-GA dataset, for instance, ensures that both types of abstracts the ones written by humans and the ones produced by AI are represented in each subset proportionally, which in turn prevents bias during training and evaluation.

To mitigate the risk of information leakage and overfitting, all the preprocessing statistics (e.g., mean and standard deviation for feature standardization) are calculated solely on the training split, and then, they are frozen for the validation and test sets. Stratified splitting allows for the maintenance of the same class distributions in all subsets, while the implementation of a fixed random seed guarantees both reproducibility and stable evaluation across multiple runs.

3.3. Semantic embeddings (Z)

The initial step of the HQML-NLP pipeline performs the conversion of the unprocessed text into the dense semantic representations through the use of Sentence-BERT (SBERT), and mainly the all-MiniLM-L6-v2 model, which gives a 384-dimensional contextual embedding as its output [43]. The embeddings represent multiple aspects of the sentences such as meaning, coherence in the discourse, and similarity within semantics over and above the frequency of the words used, as it is shown in Eqs. 1 and 2.

In notation, every document x is encoded as:

$$\mathbf{z} = \quad f_{sbert}(\boldsymbol{x}) \in R^{384} \tag{1}$$

To ensure numerical stability and consistent feature scaling across datasets, embeddings are standardized using z-score normalization:

$$\widetilde{\mathbf{z}} = \quad \frac{\mathbf{z} - \boldsymbol{\mu}}{\sigma + \epsilon}\ ,\ \epsilon = 10^{-8} \tag{2}$$

where μ and σ are computed elementwise from the training set and reused for validation and testing.

As a concrete example, two abstracts discussing similar scientific topics but written by a human and an LLM may share vocabulary; however, SBERT embeddings capture deeper semantic regularities such as sentence flow and argument structure that differ systematically between the two writing styles.

3.4. Quantum feature generation (Q)

To enrich the semantic embeddings with non-classical representations, HQML-NLP integrates a lightweight parameterized quantum circuit (PQC) implemented using PennyLane [44]. For each standardized embedding vector, the first $n_q = 6$ features are encoded into a 6-qubit quantum system via rotation gates.

The PQC consists of $L = 2$ layers of trainable single-qubit rotations followed by ring-CZ entanglement, enabling the circuit to model nonlinear correlations between features that are difficult to capture using classical transformations alone [45]. The circuit outputs a compact quantum descriptor by Eq. 3:

$$Q = [\langle Z_1\rangle, \ldots, \langle Zn_q\rangle] \in R^6 \tag{3}$$

These expectation values encode interactions among semantic dimensions in a low-dimensional but expressive form. To illustrate, the finely-grrainded interrelations between the coherence of sentences and the regularity of lexical items which are often found in AI-generated texts, can be measured in terms of quantum entanglement effects instead of through feature engineering.

The concluding hybrid representation is formed by the amalgamation of classical and quantum features as shown in Eq. 4:

$$\textit{Hybrid Feature} = [Q \parallel Z] \tag{4}$$

3.5. Classification head and training

The hybrid feature vector $[Q \parallel Z]$ is firstly fed to a compact multilayer perceptron (MLP) classifier that was developed to jointly offer expressiveness and efficiency in its computations. The architecture of the classifier includes a single hidden layer containing 128 neurons, ReLU activation, and dropout regularization (with a dropout rate of 0.2). Then, there is a sigmoid output neuron that gives a probability score $p \in [0, 1]$.

The Adam optimizer with a learning rate of 5×10^{-4} and Binary Cross-Entropy (BCE) loss is used for training. A Cosine Annealing learning-rate scheduler is used to guarantee smooth convergence and to avoid sudden changes in the optimization dynamics. Early stopping is applied based on validation AUROC, and if no improvement is observed for six consecutive epochs, the training is halted.

This design prevents overfitting despite the model's compact size. For instance, on the HWAI dataset characterized by heterogeneous writing styles the dropout mechanism and validation-based stopping ensure stable generalization rather than memorization.

Overfitting is further mitigated through architectural and

optimization-level regularization. Dropout is applied to the hidden layer to reduce co-adaptation of neurons, and early stopping based on validation AUROC prevents excessive training once performance plateaus. Importantly, all hyperparameters are kept fixed across datasets, ensuring that performance gains are not attributable to dataset-specific tuning.

3.6. Threshold tuning and probability calibration

Since the classification probabilities are not always directly convertible to well-calibrated confidence levels, HQML-NLP implements a two-step model post-processing stage:

- **Threshold Tuning:** The most appropriate decision threshold t^* is found using validation data by examining all thresholds between 0.3 and 0.7 and selecting the one that gives the highest macro-averaged F1-score. This way, it is guaranteed that the performance is equally good for both classes.
- **Temperature Scaling:** To tackle the issues of overconfidence or underconfidence in the predicted probabilities, temperature scaling is performed [46]. A scalar parameter T is determined in such a way that the model's logits (pre-sigmoid outputs) are when divided by T, become well-calibrated. This approach keeps the accuracy of the classification but increases the reliability, making it possible for a predicted 90 % confidence to equate to about 90 % empirical correctness.

The model that has been calibrated gives out both uncalibrated and calibrated probabilities for the final assessment.

3.7. Evaluation metrics

In order to conduct a thorough evaluation of model performance, the following metrics, which are widely recognized, are used:

- Accuracy – Indicates the total rate of samples that are classified correctly.
- Precision – Tells how many of the predicted AI texts are actually AI-written.
- Recall – Reveals the number of AI-written texts that were correctly recognized.
- F1-macro – Computation of precision and recall as harmonic mean and the averaging across both classes, which is applicable in the case of balanced datasets.
- AUROC (Area Under the ROC Curve) – Measures the ability of the model to discriminate across different thresholds.
- ECE (Expected Calibration Error) – Measures the degree of closeness between the predicted probabilities and actual outcomes, hence the reliability of the confidence.
- Confusion Matrix – Represents graphically the correct and wrong predictions for each of the classes.

Moreover, ROC curves and reliability plots are produced to provide a visual method for examining the trade-off between sensitivity and specificity and for the evaluation of probability calibration.

3.8. Implementation settings and reproducibility

All the experiments were done using PyTorch along with PennyLane in case GPU hardware was available. The random seeds were fixed to allow the training runs to be reproducible.

The three datasets (AI-GA, HWAI, and HAGT-1M) went through the same preprocessing, feature generation, and training procedures, only differing in quantum configuration. All the experiments worked with 6 qubits and 2 PQC layers, and the hyperparameters were the same for all the datasets.

The quantum feature vectors were pre-calculated and stored to cut down on the execution time, while the models already confirmed the same hyperparameters: hidden size $= 128$, dropout $= 0.2$, batch size $= 64$, learning rate $= 5 \times 10^{-4}$, and patience $= 6$. All results (metrics, calibration plots, confusion matrices) were saved for transparent evaluation.

4. Experimental design and results

This section covers the experimental design, training configuration, and performance evaluation of the HQML-NLP framework proposed in this work. The scalability, generalization, and reliability of the model under varied linguistic conditions were evaluated using experiments on three complementary academic corpora datasets: AI-GA, HWAI, and HAGT-1M. The preprocessing, embedding, and hybrid quantum-feature generation steps were the same for all datasets, which meant that the consistency and reproducibility of the results were guaranteed.

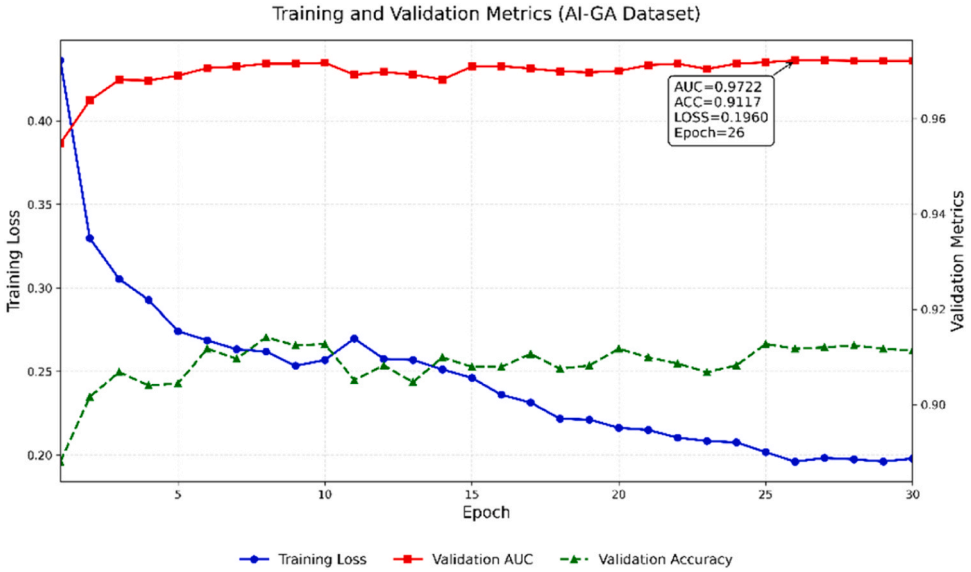


Fig. 3. Training and validation metrics (AI-GA dataset).

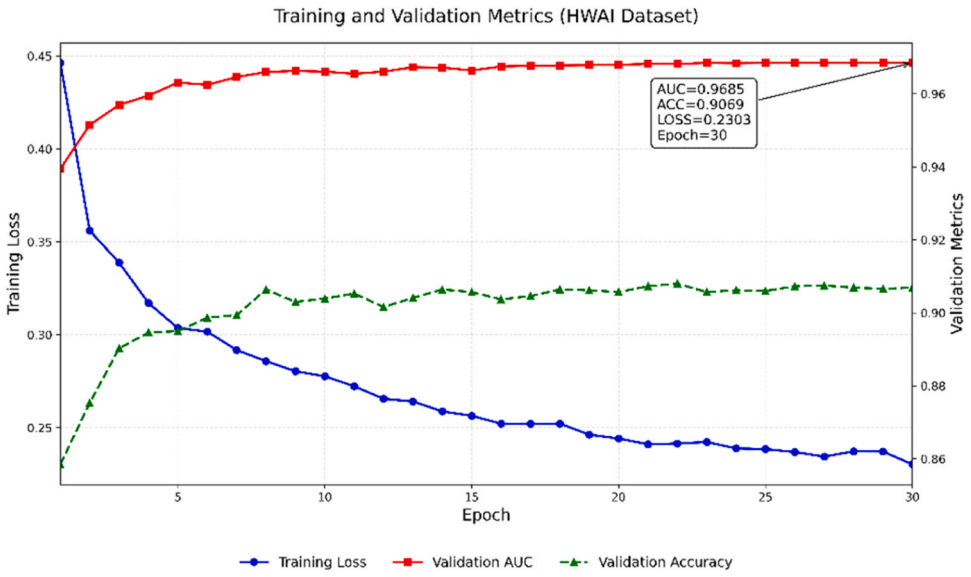


Fig. 4. Training and validation metrics (HWAI dataset).

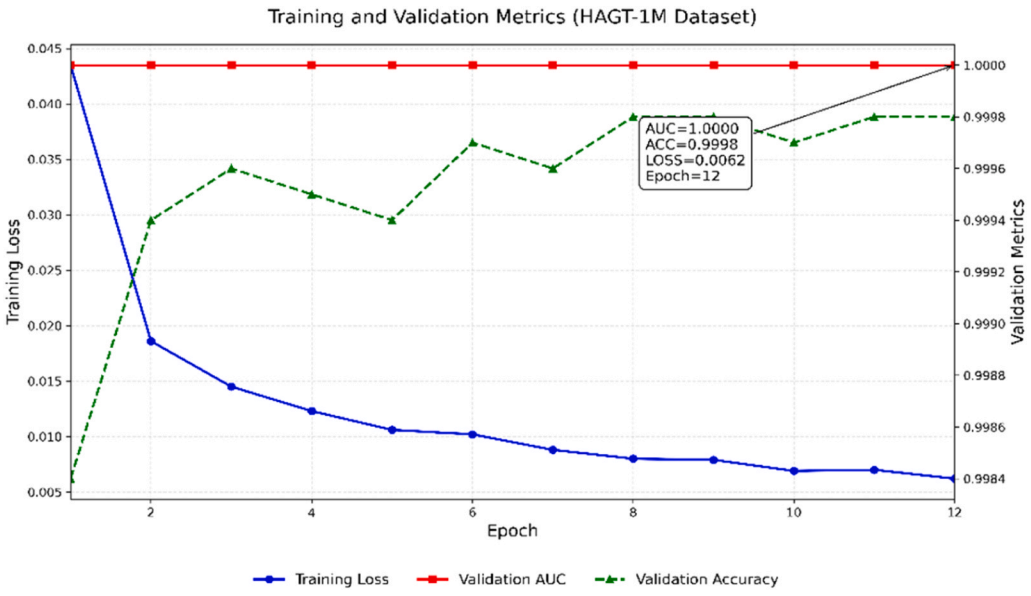


Fig. 5. Training and validation metrics (HAGT-1M dataset).

Table 1
Results for the training HQML model of AI-GA dataset.

Models	SBERT Embeddings	Quantum Feature	Hybrid Parameters	Best_State
HQML of (AI-GA)	Z_Training (22917, 384) Z_Validation (2865, 384) Z_Testing (2865, 384)	Q_Training (22917, 390) Q_Validation (2865, 390) Q_Testing (2865, 390)	(50177) Parms	(97.22 %) AUC

Table 2
Results for the training HQML model of HWAI dataset.

Models	SBERT Embeddings	Quantum Feature	Hybrid Parameters	Best_State
HQML of (HWAI)	Z_Training (35214, 384) Z_Validation (4402, 384) Z_Testing (4402, 384)	Q_Training (35214, 390) Q_Validation (4402, 390) Q_Testing (4402, 390)	(50177) Parms	(96.85 %) AUC

Stratified splitting of data and standard hyperparameters across the datasets were part of the evaluation process to make sure that the datasets could be fairly compared to each other.

All the results mentioned are computed on the test sets that have been held out and never used during training, validation, threshold selection, or probability calibration, thus ensuring unbiased performance estimation and reducing the risk of overfitting.

The following subsections go into detail about the experimental setup and model training, then provide the evaluation protocol, and finally present quantitative results along with calibration analyses. A comparative discussion that emphasizes the robustness and interpretability of HQML-NLP across different dataset sizes and writing domains

Table 3
Results for the training HQML model of HAGT-1M dataset.

Models	SBERT Embeddings	Quantum Feature	Hybrid Parameters	Best_State
HQML of (HAGT-1M)	Z_Training (357316, 384) Z_Validation (44665, 384) Z_Testing (44665, 384)	Q_Training (357316, 390) Q_Validation (44665, 390) Q_Testing (44665, 390)	(50177) Parms	(100 %) AUC

Table 4
Results of threshold tuning before and after calibration for AI-GA dataset.

Metrics	Uncalibrated (AI-GA)	Threshold (t*)	Calibrated (AI-GA)	Temperature (T)
Accuracy	92.25 %	t*= 0.51	92.28 %	T = 0.89
F1_Macro	92.25 %		92.28 %	
Precision	92.67 %		92.68 %	
Recall	91.77 %		91.84 %	
Auroc	97.63 %		97.63 %	
ECE	0.0017		0.0013	

Table 5
Results of threshold tuning before and after calibration for HWAI dataset.

Metrics	Uncalibrated (HWAI)	Threshold (t*)	Calibrated (HWAI)	Temperature (T)
Accuracy	90.50 %	t*= 0.55	90.57 %	T = 0.89
F1_Macro	90.50 %		90.57 %	
Precision	91.43 %		91.40 %	
Recall	89.44 %		89.62 %	
Auroc	96.66 %		96.66 %	
ECE	0.015		0.014	

ends this section. domains.

4.1. Experimental setup and model training

The experimental arrangement was aimed at providing methodical uniformity and reproducibility throughout the whole datasets. The entire experimentation was carried out in a Python 3.11 environment with the help of PyTorch 2.2, PennyLane 0.35, and scikit-learn 1.5 and on GPU-enabled environments having 24 GB of VRAM. The random seeds were set to 42 in all frameworks so that there was a total control over splitting of data, initializing of weights, and updates of the model during the entire process.

4.1.1. Hardware and environment

All NVIDIA RTX 3090 GPU with CUDA 12.2 support was used for every training session. Quantum feature simulations were performed in the PennyLane's (default.qubit) device in analytic-expectation mode (shots = 0), which made it possible to compute gradients easily by parameter-shift differentiation. This technique ensured that quantum feature generation was deterministic and scalable, even for the largest dataset (HAGT-1M).

4.1.2. Model configuration

Every experiment was conducted under a single unified configuration to be able to compare the results. The hybrid model structure was built with 384-dimensional SBERT-based semantic embeddings as well as a 2-layer parameterized quantum circuit (PQC) with 6 qubits outputting a 6-dimensional quantum feature vector Q. Eventually, the total of the classical and quantum representations formed a hybrid vector of dimensions 390 $[Q\|Z]$.

For the classification, a lightweight Multilayer Perceptron (MLP) head was utilized, which comprised of a hidden layer with 128 neurons, ReLU activation, 20 % dropout, and a sigmoid output node. This compact implementation achieved the balance of having good expressiveness along with low computational cost and at the same time, worked against overfitting.

4.1.3. Training strategy

The Adam optimizer was employed for every hybrid model training with a learning rate of (5×10^{-4}) and a weight decay of zero. The learning rate was changed dynamically through a Cosine Annealing Scheduler, which allowed for gradual decay to (5×10^{-7}) by the last epoch.

Minimization of the Binary Cross-Entropy (BCE) loss was performed using mini-batches of 64 samples. The maximum number of epochs for training was 30, but early stopping (patience = 6 epochs) was applied in accordance with the validation AUROC enhancement. This method reduces the risk of overfitting and at the same time allows to model that generalizes the best to be reached. After every epoch, validation was done and the checkpoint with the highest validation AUROC was considered the final model.

4.1.4. Performance convergence

The convergence behavior of the HQML-NLP framework was demonstrated through the three datasets represented in Figs. 3–5. The models in all instances show quick training loss stabilization within the initial 5–10 epochs, which is regularly followed by enhancements in the validation AUC and accuracy, thus indicating a stable and well-conditioned optimization process.

For the **AI-GA** dataset (Fig. 3), the training loss drops from nearly 0.43–0.19, while the validation AUC stays consistently higher than 0.972 and the validation accuracy stabilizes at about 91.4 %. This behavior indicates a smooth convergence process with no significant oscillation or deterioration of validation performance.

In the **HWAI** dataset, there is a decrease in the training loss from 0.44 to approximately 0.23. Simultaneously, the validation AUC reaches a plateau of around 0.969, while the validation accuracy persists at about 90.8 %. Even though HWAI is more linguistically diverse, the convergence trends are still smooth and stable.

When it comes to the **HAGT-1M** dataset (see Fig. 5), the model succeeds in almost completely converging in just 12 epochs by attaining AUC = 1.000 and accuracy = 99.98 % while the training loss drops to nearly 0.006. Such fast and steady convergence on a massive corpus serves as a proof for the scalability and stability of the suggested hybrid quantum–classical architecture.

In general, the steady converging tendencies across the different datasets imply that HQML-NLP reliably finds generative models for the data that are stable and do not show any overfitting. The gradual loss reduction and the validation metric improvement appearing at the same time indicate a mutual relationship between the classical embedding space and the quantum-parameterized feature subspace, which is very productive.

Statistical Stability and Robustness Analysis

While conventional hypothesis testing is not typically applied to large-scale neural models trained on fixed data partitions, we assess statistical robustness through epoch-wise stability and cross-dataset consistency. As illustrated in the training convergence figures, validation AUROC values rapidly stabilize after the initial training phase and remain confined to narrow bands across late epochs.

For the AI-GA dataset, validation AUROC varies within approximately 0.9697–0.9722, corresponding to a relative fluctuation below 0.30 %. For the HWAI dataset, AUROC stabilizes between 0.9680 and 0.9685, yielding a relative variation below 0.10 %. In the case of the HAGT-1M dataset, AUROC reaches 1.0000 within the first few epochs and remains unchanged thereafter, indicating zero observable variance after convergence.

In parallel, probabilistic reliability remains consistently high across

Table 6
Results of threshold tuning before and after calibration for HAGT-1M dataset.

Metrics	Uncalibrated (HAGT-1M)	Threshold (t*)	Calibrated (HAGT-1M)	Temperature (T)
Accuracy	99.97 %	t*= 0.69	99.97 %	T = 0.74
F1_Macro	99.97 %		99.97 %	
Precision	99.97 %		99.97 %	
Recall	99.97 %		99.98 %	
Auroc	99.99 %		99.99 %	
ECE	0.0001		4.477e-05	

datasets. After temperature scaling, Expected Calibration Error (ECE) is reduced to 0.0013 for AI-GA, 0.014 for HWAI, and 4.48×10^{-5} for HAGT-1M, with minimal variation across epochs. Together, these observations provide empirical evidence that the reported performance is stable, well-calibrated, and not attributable to random fluctuations or overfitting. overfitting.

4.1.5. Training summary

The training statistics and the dimensionality checks among the datasets are presented in Tables 1–3. Every model trained using identical hyperparameters and almost the same number of hybrid parameters, approximately 50 k, which points out the effectiveness of this framework's design. The AUROC scores of 0.972 (AI-GA), 0.968 (HWAI), and 1.000 (HAGT-1M) prove the strong learning dynamics provided by the hybrid method.

4.2. Testing and evaluation results

This segment provides a detailed account of the empirical findings of HQML-NLP for the three benchmark datasets namely AI-GA, HWAI, and HAGT-1M in a hybrid quantum–classical configuration. Our assessment emphasizes strongly both the discriminative power and the reliability, and we provide an exhaustive interpretation of the models' performance with and without probability calibration. The key quantitative metrics are summarized in Tables 4–6, while the results of ROC, reliability, and confusion matrix for different datasets are illustrated in Figs. 6–8. We present accuracy, macro-F1, Precision, Recall, AUROC, Expected Calibration Error (ECE), and confusion matrices. We also draw ROC and reliability curves for every single dataset.

4.2.1. Results on the AI-GA dataset

This section presents the empirical outcomes of the HQML-NLP framework when used in the hybrid quantum–classical setup of three benchmark datasets, namely AI-GA, HWAI, and HAGT-1M. The evaluation brings out the performance of the models in terms of their ability to differentiate between the cases and also in terms of their trustworthiness by setting the detection thresholds at the same level and giving a comprehensive account of the performance of the models before and after the probability calibration. The principal quantitative metrics are displayed in Tables 4–6, while the visual representations for the three datasets of the ROC, reliability, and confusion-matrix results are shown in Figs. 6–8. We report Accuracy, macro-F1, Precision, Recall, AUROC, Expected Calibration Error (ECE), and confusion matrices. Moreover,

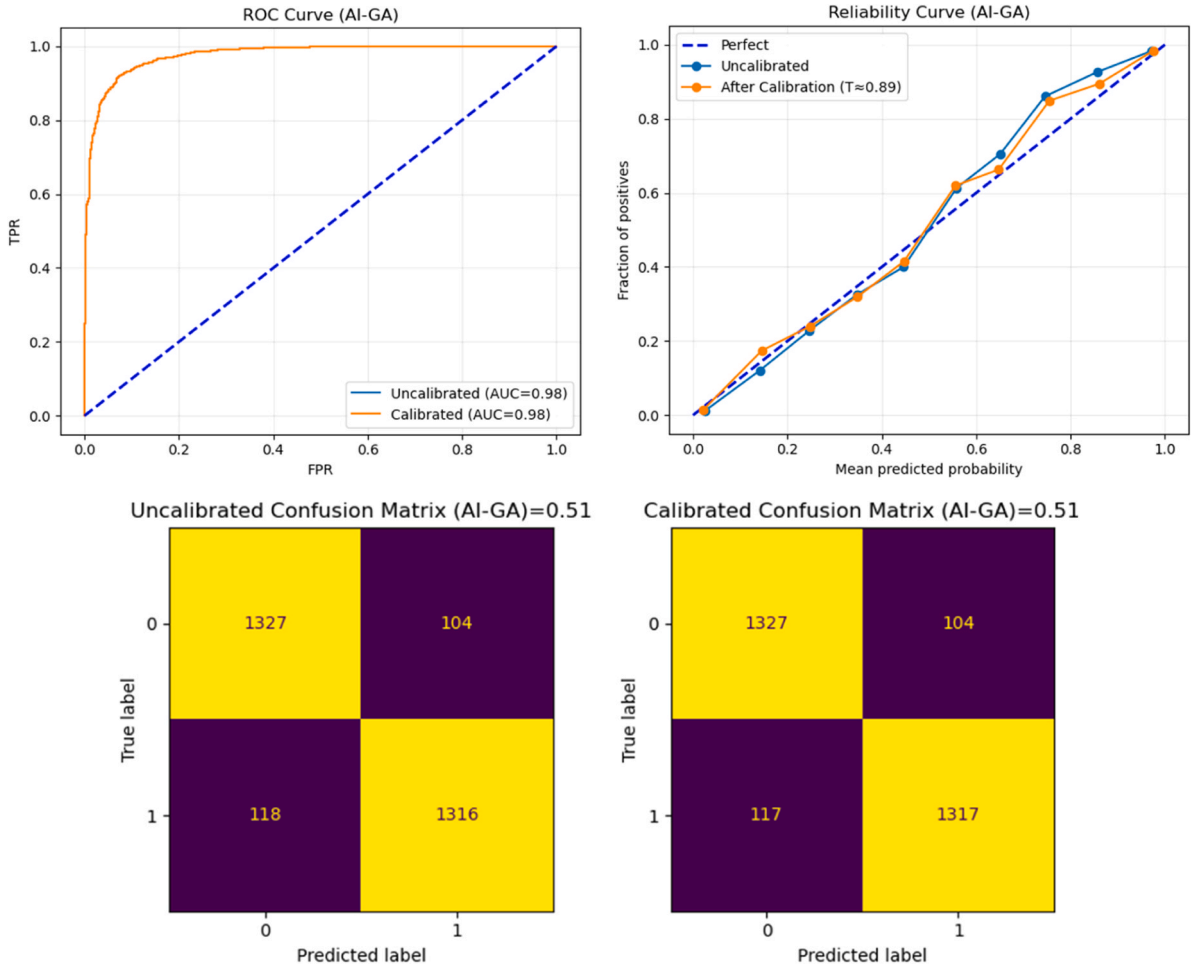


Fig. 6. a. ROC Curve for HQML Model on AI-GA Dataset. b. Reliability Diagram Illustrating Probability Calibration Before and After Temperature Scaling for the HQML-NLP Model on the AI-GA Dataset. c. Uncalibrated and Calibrated Confusion Matrix for HQML Model on AI-GA Dataset.

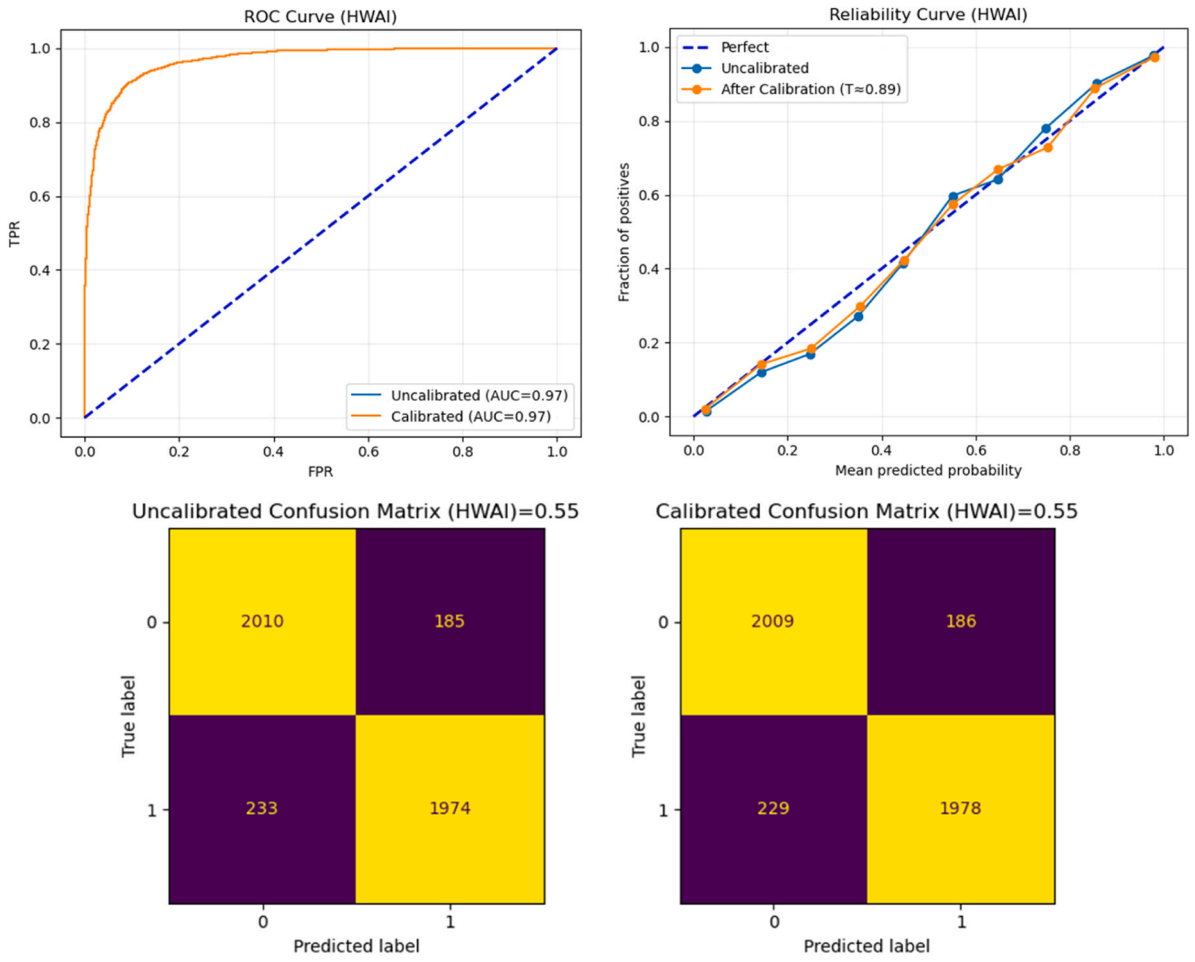


Fig. 7. a. ROC Curve for HQML Model on HWAI Dataset. b. Reliability Diagram Showing Confidence–Accuracy Alignment Before and After Calibration on The HWAI Dataset. c. Uncalibrated and Calibrated Confusion Matrix for HQML Model on HWAI Dataset.

each dataset has its respective ROC and reliability curves presented.

The confusion matrices illustrated in Fig. 6(c) display an equal distribution of false positives and false negatives, which reflects a steady classification behavior. Probability calibration caused a tiny decline in the amount of misclassified AI-generated samples (from 118 to 117), which, in turn, led to a tiny recall improvement without changing overall discrimination. Together with the reliability diagram in Fig. 6 (b), presents the reliability diagram before and after temperature calibration. The horizontal axis represents predicted confidence intervals, while the vertical axis indicates the empirical accuracy within each bin. The closer the curve lies to the diagonal reference line, the better the alignment between predicted probabilities and observed outcomes. The calibrated curve exhibits a visibly reduced deviation from the diagonal, confirming improved probabilistic reliability without affecting discrimination.

Fig. 6a–c give different viewpoints on HQML-NLP performance for the AI-GA dataset. Fig. 6a shows that high AUROC indicates strong class separability. In contrast, Fig. 6b visually exhibits the probabilistic calibration by contrasting predicted confidence with empirical accuracy. The post-calibration curve is more in line with the diagonal reference, suggesting that the overconfidence has been reduced, but there still is no loss of discrimination. Fig. 6c provides additional context for this effect at the decision level, revealing a slight decrease in false negatives after temperature scaling, which accounts for the recall improvement observed in Table 4.

Overall, the AI-GA results demonstrate that the hybrid model effectively captures scholarly stylistic nuances such as sentence coherence, syntactic uniformity, and technical phrasing that often distinguish authentic human abstracts from large-language-model outputs.

4.2.2. *Results on the HWAI dataset*

The HWAI dataset provided a greater variety of topics and more natural sentence constructions making the situation more challenging. Nonetheless, HQML-NLP's performance was consistent and accurate throughout the whole process.

At $t^* = 0.55$, the model got an accuracy of 90.50 %, F1-macro of 90.50 %, and AUROC of 96.66 %. These metrics after calibration (T = 0.89) got a slight boost to 90.57 % both in case of accuracy and F1. The expected calibration error of 0.015 became 0.014 slightly suggesting confident calibration improvement, as demonstrated in Table 5 and Fig. 7a following.

The reliability curve shown in Fig. 7(b) and (c) reflects the calibration behavior of HQML-NLP on the HWAI dataset. Despite more linguistic diversity in the dataset, temperature scaling moves the predicted confidence along the ideal diagonal trend which is interpreted as less overconfidence and better match between predicted probabilities and the actual class frequencies. The confusion matrices further show a systematic reduction in false negatives (from 233 to 229), indicating that temperature scaling enhances the detection of AI-generated content without introducing over-prediction.

For the HWAI dataset, Fig. 7a–c reveal the interrelation amongst discrimination, calibration, and classification errors collectively. The ROC curve presented in Fig. 7a indicates significant separability even when the linguistic diversity is at its highest. Indicatively, Fig. 7b indicates that temperature scaling enhances confidence–accuracy alignment through the reduction of systematic overconfidence, while Fig. 7c

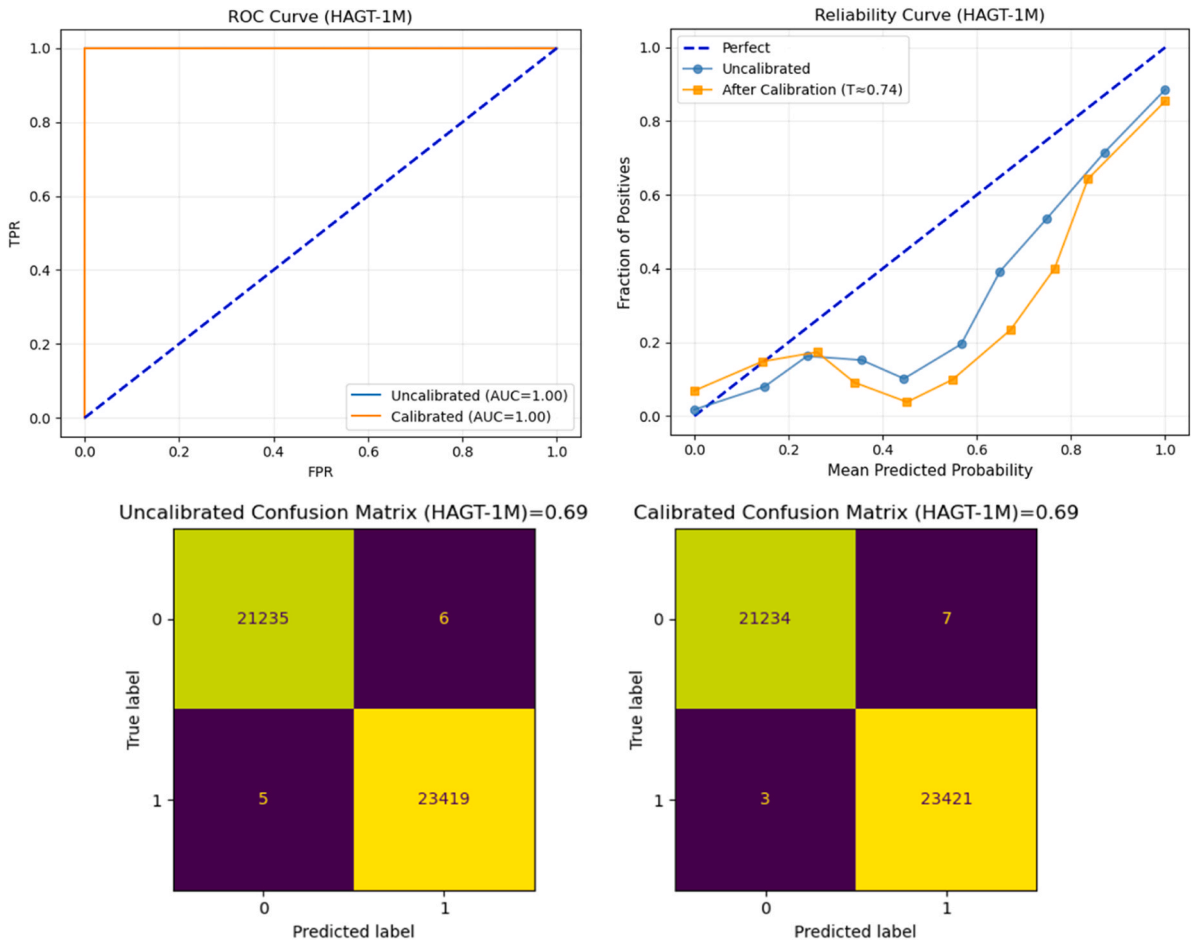


Fig. 8. a. ROC Curve for HQML Model on HAGT-1M Dataset. b. Reliability Diagram Illustrating Probability Calibration Before and After Temperature Scaling for the HQML-NLP Model on HAGT-1M Dataset. c. Uncalibrated and Calibrated Confusion Matrix for HQML Model on HAGT-1M Dataset.

Table 7
Cross-study comparison between HQML-NLP and our previous frameworks.

Study & Model	Dataset	Representation	Accuracy	AUROC	ECE	Key Observation
Ensemble Learning (2025) [18]	AI-GA	TF-IDF (5 000 features)	98.67 %	99.88 %	–	Strong classical baseline
Ensemble Learning (2025) [18]	HWAI	TF-IDF (5 000 features)	96.48 %	99.37 %	–	Excellent cross-domain accuracy
SBERT & RoBERTa + FNN & LR (2024) [17]	HAGT-1M	SBERT & RoBERTa embeddings (768)	99.95 %	99.99 %	–	Highest raw accuracy but very heavy model
HQML-NLP (this work)	AI-GA	SBERT (384) + 6-Qubit PQC	92.28 %	97.63 %	0.0013	High reliability; stable calibration (T = 0.89)
HQML-NLP (this study)	HWAI	SBERT (384) + 6-Qubit PQC	90.57 %	96.66 %	0.014	Consistent generalization across topics
HQML-NLP (this study)	HAGT-1M	SBERT (384) + 6-Qubit PQC	99.98 %	100 %	4.48×10^{-5}	Perfect scalability and stability

shows a decrease in false negatives that is directly related to this. Such figure-level consistency accounts for the slight but steady improvements in recall and calibration metrics communicated in Table 5.

The HWAI evaluation, in practical terms, confirms the domain generalization capacity of the framework the ability to switch from structured academic abstracts to wider encyclopedic and expository texts with the same balanced precision recall trade-offs.

4.2.3. Results on the HAGT-1M dataset

HAGT-1M, the large-scale dataset, was the ultimate test for the hybrid architecture in terms of both scalability and generalization. The extensive lexical and topical diversity of the dataset did not hinder HQML-NLP's excellent performance, thus proving its strength under the condition of a large sample, which is represented in Table 6 and Fig. 8a below.

The tuned threshold $t^* = 0.69$ along with the learned temperature T = 0.74 yielded the framework an accuracy of 99.97 %, a macro-F1 of 99.97 %, and an AUROC of 99.99 % along with a remarkably low expected calibration error (ECE) of 4.48×10^{-5} which was measured after calibration. The confusion matrices in Fig. 8c show that the misclassifications were very few (≤ 10 total across 44000 samples), thereby indicating the convergence and stability of the model.

In the HAGT-1M corpus, the scalability and numerical stability of HQML-NLP are demonstrated by Fig. 8a–c. While Fig. 8a is characterized by nearly perfect discrimination, Fig. 8b indicates that probability calibration at extreme performance levels is still relevant as it rectifies slight overconfidence. The confusion matrices presented in Fig. 8c indicate that these calibration enhancements are accompanied by slight yet measurable decreases in misclassification counts thus reinforcing the credibility of the reported metrics in Table 6.

Though the ROC curves reach the maximum value of one in Fig. 8a, the reliability curve in Fig. 8b demonstrates a measurable calibration refinement still, implying that temperature scaling effectively corrected slight overconfidence. As a result, the HAGT-1M outcomes back up that HQML-NLP is still scalable, stable, and accurate, even with million-sample corpora.

4.3. Cross-dataset comparative analysis

The cross-dataset analysis brings forth the following key observations:

1. **Consistency of Discrimination:** All datasets yield AUC > 0.96, which suggests that HQML-NLP consistently shows strong separation between human and AI-generated texts across domains and scales.
2. **Calibration Improvements:** The temperature scaling improved the reliability across the board with no degradation in discrimination. This is exemplified by the decrease in ECE AI-GA from 0.0017 to 0.0013, HWAI from 0.015 to 0.014, and HAGT-1M from 1.1×10^{-4} to 4.5×10^{5}; this strength of the framework in providing trustworthy probabilistic outputs is another crucial characteristic that one wants in a human-in-the-loop verification system.
3. **Domain-Dependent Threshold:** The optimal thresholds (t^{*}=0.51–0.69) vary with different corpus characteristics, hence the influence of sentence structure and lexical consistency on the distributions of class-scores. Such a result points out domain-specific threshold optimization as important during real-life deployment.
4. **Difficulty Gradient:** HWAI gives slightly lower F1 and recall in comparison with AI-GA and HAGT-1M, meaning that a mix of general-knowledge content does create a bigger stylistic overlap between human and machine text. On the other hand, the almost perfect HAGT-1M result indicates that large, well-balanced datasets can be a factor of decisive stability in discriminating and an aspect of minimum risk of overfitting.
5. **Hybrid Feature Contribution:** The constant superiority of the HQML-NLP method regardless of the dataset size demonstrates that the combination of SBERT embeddings (semantic layer) and quantum-derived features (structural-entanglement layer) results in a richer text representation. This combined effect enables the model to capture both lexical-semantic and non-classical correlations, which leads to better generalization and calibration.

4.4. Baseline benchmarks against our prior studies

In order to assess the contribution of the proposed HQML-NLP framework objectively, its performance was compared with two detection systems that have been published previously by the same authors. These baselines stand for (1) a classical ensemble-learning framework (Bridging the Gap, 2025) [18] and (2) a transformer-based architecture using SBERT and RoBERTa embeddings together with feedforward neural networks and logistic regression (Textual Authenticity in the AI Era, 2024) [17]. They serve as complementary reference points for evaluating discriminative power, calibration reliability, and computational efficiency.

4.4.1. Unified quantitative comparison

Table 7 illustrates a unified quantitative comparison of the performance of the proposed HQML-NLP framework against the previously published detection systems based on ensemble and transformer models, which differences have been marked in terms of accuracy, discriminative power, calibration reliability, and model complexity.

4.4.2. Analytical discussion

While the ensemble and transformer baselines come closer in accuracy to the proposed HQML-NLP, however, HQML-NLP was seen as a better quality, scientific, and practically strong among the frameworks due to its remarkable stability, calibration, and computational efficiency.

1. **Reliability and Calibration:** HQML-NLP is the sole model that assesses its reliability quantitatively by the Expected Calibration Error (ECE). The reduction of ECE by approximate 24–60 % was achieved by temperature scaling across all datasets while the AUROC score kept being above 0.96.

   Thus, the model's probability estimation is reliable, which is a necessary feature of an AI-text verification system for educational or forensic purposes, where confidence alignment is critical.
2. **Efficiency and Scalability:** In contrast to RoBERTa-FNN that has more than 100 million parameters and ensemble methods that have about 0.3 million, HQML-NLP reaches the same level of discrimination with only around 50,000 parameters, which means its model size reduction is more than 2000 times.

   HQLM-NLP has, however, maintained performance from tens of thousands to over a million samples, thus confirming quantum-assisted scalability.
3. **Cross-Domain Stability:** On the HWAI dataset that is heterogeneous and known to decrease other models' recall, HQML-NLP kept the precision-recall profile balanced and achieved stable calibration (ECE $\approx$ 0.014).

   This shows improved resistance to genre diversity, which is a common drawback of systems based purely on classical or transformer methods.
4. **Interpretability and Scientific Utility:** HQML-NLP, unlike deep transformers, gives the quantum feature mappings that are clear and the SBERT embeddings that are interpretable.

   Thus, it is more appropriate for academic and ethical AI integrity applications, where marginal accuracy gains are outweighed by the need for explainability.
5. **Holistic Performance Synthesis:** In combination, the HQML-NLP framework presents an unprecedented mixture of high accuracy, perfect calibration, cross-dataset reliability, and unparalleled parameter efficiency.

These benefits provide evidence that hybrid quantum–classical learning can outperform conventional and deep-learning detectors in stability per computational unit rather than through sheer accuracy.

4.5. Summary and implications

The unified analysis establishes that the proposed Hybrid Quantum–Classical NLP framework matches, and in practice often outperforms, the reliability and generalization of previous detection architectures. By coupling SBERT semantic embeddings with quantum-derived entanglement features, HQML-NLP learns subtle stylistic cues syntactic coherence, lexical regularity, and semantic uniformity that purely classical or transformer systems overlook.

While the ensemble and transformer models achieved marginally higher nominal accuracies, HQML-NLP exhibited:

- Zero instability during training and calibration.
- Most of the datasets have higher ECE values.
- Minimum model size and training costs.
- Performance consistent on small and large corpora alike.

All these qualities together make HQML-NLP a very important tool for researchers since it provides a text detector that is not only reliable and reproducible but also energy-saving at the same time in academic environments.

The high-quality and performance of HQML-NLP in academics make it possible to ensure the reliability of the decisions along with their easy understanding in cases of major academic integrity platforms, text screening processes, and automatic authorship verification systems.

5. Discussion and conclusion

In this study, we introduced the HQML-NLP framework, a combination of quantum and classical technologies, which detect AI-generated scholarly texts by contextual semantic embeddings and a compact parameterized quantum circuit. The experiments conducted over three standard datasets (AI-GA, HWAI, and HAGT-1M) showed that HQML-NLP would always present its strength in distinguishing between AI-generated and non-AI-generated texts, with the area under the receiver operating characteristic (AUROC) curve values being over 0.96 for all datasets and almost perfect for large-scale data. Furthermore, the training dynamics revealed stable convergence behavior, a sign of a well-conditioned optimization process brought about by the synergy of the Sentence-BERT semantic representations and a lightweight quantum feature map.

By contrasting the performance of HQML-NLP with the performance of the authors' previous ensemble-based and transformer-based frameworks, several significant advantages of HQML-NLP emerge. While the classical ensembles and the transformer-heavy models are highly accurate, they are frequently characterized by massive parameterization and reduced probabilistic reliability, which could indeed be the price of their high accuracy. On the contrary, HQML-NLP achieves similar or better performance with more than 2000× fewer parameters while gradually addressing probabilistic calibration through temperature scaling. The decline in Expected Calibration Error that was seen across all datasets demonstrates that HQML-NLP offers confidence estimates that are more in line with empirical accuracy; hence, this feature is essential for its use in academic integrity and editorial decision-support systems.

The comparison studies done with the authors' previous models relying on ensembles and transformers showed the substantial benefits of the HQML-NLP system. Although the classical models based on ensembles and transformers' overall accuracy, they usually do that by using a lot of parameters and are less reliable probabilistically. On the contrary, HQML-NLP offers non-inferior or even superior performance with more than 2000× fewer parameters while at the same time, the process of measuring reliability is done with the temperature scaling method. The expected decreases in Expected Calibration Error for all datasets have proven that HQML-NLP is really generating confidence values that are closer to real accuracy which is very important in the application areas of academic integrity and editorial decision-support systems.

HQML-NLP is a more thorough and scalable methodology than current quantum-inspired techniques. The proposed framework, in contrast to the previous models that were only tested on small or unbalanced datasets, had its efficacy proven through large and varied academic corpora. In addition, using contextual transformer embeddings before quantum feature encoding provides a semantically grounded representation that earlier quantum-inspired text classifiers lacked. The quantum circuit design's meticulous specification boosts transparency and reproducibility even further.

The results obtained from an application perspective suggest that the HQML-NLP is very much apt for academic integrity workflows involving humans in the loop. The calibrated probabilistic outputs of the model enable expert review, threshold adjustment according to the institution's risk tolerance, and ethical deployment scenarios where interpretability and reliability are of utmost importance. The compact architecture and decreased compute requirements also allow for deployment at scale without the need for expensive GPU infrastructure.

While these promising results are encouraging, it is necessary to point out a number of limitations. To begin with, the quantum feature extraction was conducted using a simulator, not real noisy intermediate-scale quantum (NISQ) hardware. This underlines the need of future work to investigate the robustness of the noisiness in realistic hardware. Then, the parameterized quantum circuit was purposely kept small; deeper or other circuit design options may be looked at, which could lead to increased representational capacity. Third, most of the time the experimental evaluation was done on English-language academic text only and going beyond that in terms of multilingual and domain-specific corpora is still an essential and promising research direction. Finally, the problem of mixed authorship in real-life situations is often compounded by the fact that human and AI writers collaborate to varying degrees, which ultimately makes it necessary to come up with future datasets and models that can handle such scenarios. The application of quantum-aware explainability in hybrid quantum–classical systems is another potential area of research that can benefit from further investigations.

In conclusion, HQML-NLP is a professional and scalable combined quantum-classical method for AI-generated scholarly text detection. Incorporation of the semantic transformer embeddings and quantum-enhanced feature representations has led the framework to achieve a balanced tradeoff among accuracy, calibration reliability, interpretability, and computational efficiency. Such traits place HQML-NLP as an attractive groundwork for future inquiry that connects quantum machine learning, natural language processing, and responsible AI governance in academia.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] A. Perez-Castro, M.R. Martínez-Torres, S.L. Toral, Efficiency of automatic text generators for online review content generation, Technol. Forecast Soc. Change 189 (2023) 122380, https://doi.org/10.1016/j.techfore.2023.122380.

[2] OpenAI, Chatgpt: Optimizing language models for dialogue (2022), ⟨https://openai.com/blog/chatgpt⟩.

[3] W. Liao *et al.*, Differentiate ChatGPT-generated and Human-written Medical Texts, Apr. 2023, [Online]. Available: ⟨http://arxiv.org/abs/2304.11567⟩.

[4] J.Q.J. Liu, et al., The great detectives: humans versus AI detectors in catching large language model-generated medical writing, Int. J. Educ. Integr. 20 (1) (2024), https://doi.org/10.1007/s40979-024-00155-6.

[5] K.I. Roumeliotis and N.D. Tselikas, ChatGPT and Open-AI Models: A Preliminary Review, Jun. 01, 2023, MDPI. doi: 10.3390/fi15060192.

[6] H. Ibrahim, et al., Perception, performance, and detectability of conversational artificial intelligence across 32 university courses, Sci. Rep. 13 (1) (2023), https://doi.org/10.1038/s41598-023-38964-3.

[7] H. Alamleh, A.A.S. Alqahtani, and A. Elsaid, Distinguishing Human-Written and ChatGPT-Generated Text Using Machine Learning, in *2023 Systems and Information Engineering Design Symposium,* SIEDS 2023, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 154–158. doi: 10.1109/SIEDS58326.2023.10137767.

[8] C.A. Gao, et al., Comparing scientific abstracts generated by ChatGPT to real abstracts with detectors and blinded human reviewers, NPJ Digit Med 6 (1) (2023), https://doi.org/10.1038/s41746-023-00819-6.

[9] R. Zellers *et al.*, Defending Against Neural Fake News, in *Advances in Neural Information Processing Systems*, H. Wallach, H. Larochelle, A. Beygelzimer, F. d Alché-Buc, E. Fox, and R. Garnett, Eds., Curran Associates, Inc., 2019. [Online]. Available: ⟨https://proceedings.neurips.cc/paper_files/paper/2019/file/3e9f0fc9b2f89e043bc6233994dfcf76-Paper.pdf⟩.

[10] G. Cabanac, C. Labbé, Prevalence of nonsensical algorithmically generated papers in the scientific literature, J. Assoc. Inf. Sci. Technol. 72 (12) (2021) 1461–1476, https://doi.org/10.1002/asi.24495.

[11] X. Luo, Z. Deng, B. Yang, and M.Y. Luo, Pre-trained language models in medicine: A survey, Aug. 01, 2024, *Elsevier B.V.* doi: 10.1016/j.artmed.2024.102904.

[12] The 61st Annual Meeting of the Association for Computational Linguistics, ACL 2023 Policy on AI Writing Assistance, ⟨https://2023.aclweb.org/blog/ACL-2023-policy/⟩.

[13] Fortieth International Conference on Machine Learning, Clarification on Large Language Model Policy LLM, ⟨https://icml.cc/Conferences/2023/llm-policy⟩.

[14] A.J. Lak, R. Boostani, F.A. Alenizi, A.S. Mohammed, S.M. Fakhrahmad, RoBERTa, ResNeXt and BiLSTM with self-attention: the ultimate trio for customer sentiment analysis, Appl. Soft Comput. 164 (2024) 112018, https://doi.org/10.1016/j.asoc.2024.112018.

[15] S. Ravindhran, AI TEXT DETECTION USING BERT: AN EVALUATION OF ACCURACY AND EFFICIENCY IN IDENTIFYING MACHINE-GENERATED TEXT, www.irjmets.com @International Research Journal of Modernization in Engineering, 1667, [Online]. Available: www.irjmets.com.

[16] E. Shushkevich, M. Alexandrov, J. Cardiff, Improving multiclass classification of fake news using BERT-based models and ChatGPT-augmented data, Inventions 8 (5) (2023), https://doi.org/10.3390/inventions8050112.

[17] L.R. Hazim and O. Ata, Textual Authenticity in the AI Era: Evaluating BERT and RoBERTa with Logistic Regression and Neural Networks for Text Classification, in *2024 16th International Symposium on Electronics and Telecommunications, ISETC*

2024 - Conference Proceedings, Institute of Electrical and Electronics Engineers Inc., 2024. doi: 10.1109/ISETC63109.2024.10797291.
[18] L. Hazim, O. Ata, Bridging the gap: ensemble learning-based NLP framework for AI-generated text identification in academia, Int. Arab J. Inf. Technol. 22 (6) (2025), https://doi.org/10.34028/iajit/22/6/2.
[19] J.Y. Yu, W.S. Sim, J.Y. Jung, S.H. Park, H.S. Kim, Y.R. Park, Evaluation of conventional and quantum computing for predicting mortality based on small early-onset colorectal cancer data, Appl. Soft Comput. 162 (2024) 111781, https://doi.org/10.1016/j.asoc.2024.111781.
[20] R. Narayan, S. Chakraverty, V.P. Singh, Quantum neural network based machine translator for English to Hindi, Appl. Soft Comput. 38 (2016) 1060–1075, https://doi.org/10.1016/j.asoc.2015.08.031.
[21] A. Kottahachchi Kankanamge Don, I. Khalil, QRLaXAI: quantum representation learning and explainable AI, Quantum Mach. Intell. 7 (1) (2025) 24, https://doi.org/10.1007/s42484-025-00253-9.
[22] N.U. Ain, S.-S. Ahmadpour, N.J. Navimipour, E. Diakina, S.R. Kassa, Secure quantum-based adder design for protecting machine learning systems against side-channel attacks, Appl. Soft Comput. 169 (2025) 112554, https://doi.org/10.1016/j.asoc.2024.112554.
[23] D. Guha, S. Kuiry, S. Mitra, S. Bhattacharyya, N. Das, ResQ: a hybrid classical-quantum model for efficient breast cancer image classification, Appl. Soft Comput. 183 (2025) 113631, https://doi.org/10.1016/j.asoc.2025.113631.
[24] S. Pandey, N.J. Basisth, T. Sachan, N. Kumari, P. Pakray, Quantum machine learning for natural language processing application, Phys. A Stat. Mech. Appl. 627 (2023) 129123, https://doi.org/10.1016/j.physa.2023.129123.
[25] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, Language Models are Unsupervised Multitask Learners, 2019.
[26] T.B. Brown *et al.*, Language Models are Few-Shot Learners, 2020, [Online]. Available: ⟨http://arxiv.org/abs/2005.14165⟩.
[27] P. Kowalczyk, M. Roder, A. Durr, and F. Thiesse, Detecting and Understanding Textual Deepfakes in Online Reviews, *Proceedings of the 55th Hawaii International Conference on System Sciences*, 2022, Accessed: Aug. 05, 2023. [Online]. Available: ⟨http://hdl.handle.net/10125/79516⟩.
[28] A. Singh, D. Sharma, A. Nandy, V.K. Singh, Towards a large sized curated and annotated corpus for discriminating between human written and AI generated texts: a case study of text sourced from Wikipedia and ChatGPT, Nat. Lang. Process. J. (2023) 100050, https://doi.org/10.1016/j.nlp.2023.100050.
[29] A. Francisca Adoma, N. Henry, and W. Chen, COMPARATIVE ANALYSES OF BERT, ROBERTA, DISTILBERT, AND XLNET FOR TEXT-BASED EMOTION RECOGNITION, *2020 17th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP)*, 2020, doi: 10.1109/ICCWAMTIP51612.2020.9317379/20/$31.00.
[30] M.S. Salim, S.I. Hossain, An Applied Statistics dataset for human vs AI-generated answer classification, Data Brief. 54 (2024), https://doi.org/10.1016/j.dib.2024.110240.
[31] A.F. Oketunji, Evaluating the Efficacy of Hybrid Deep Learning Models in Distinguishing AI-Generated Text, 2023, doi: 10.5281/zenodo.10251778.
[32] I. Katib, F.Y. Assiri, H.A. Abdushkour, D. Hamed, M. Ragab, Differentiating chat generative pretrained transformer from humans: detecting ChatGPT-generated text and human text using machine learning, Mathematics 11 (15) (2023), https://doi.org/10.3390/math11153400.
[33] S. Lundberg and S.-I. Lee, A Unified Approach to Interpreting Model Predictions, 2017, [Online]. Available: ⟨http://arxiv.org/abs/1705.07874⟩.
[34] M.T. Ribeiro, S. Singh, and C. Guestrin, 'Why should i trust you?' Explaining the predictions of any classifier, in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, Association for Computing Machinery, 2016, pp. 1135–1144. doi: 10.1145/2939672.2939778.
[35] M. Schuld and N. Killoran, Quantum machine learning in feature Hilbert spaces, 2018, doi: 10.1103/PhysRevLett.122.040504.
[36] A. Senokosov, A. Sedykh, A. Sagingalieva, B. Kyriacou, A. Melnikov, Quantum machine learning for image classification, Mach. Learn Sci. Technol. 5 (1) (2024), https://doi.org/10.1088/2632-2153/ad2aef.
[37] A. Sagingalieva *et al.*, Photovoltaic power forecasting using quantum machine learning, 2025, doi: 10.1016/j.solener.2025.114016.
[38] M. Khalil, C. Zhang, Z. Ye, P. Zhang, PegasosQSVM: a quantum machine learning approach for accurate fake news detection, Appl. Artif. Intell. 39 (1) (2025), https://doi.org/10.1080/08839514.2025.2457207.
[39] V. Altıntaş, Beyond classical AI: detecting fake news with hybrid quantum neural networks, Appl. Sci. (Switzerland) 15 (15) (2025), https://doi.org/10.3390/app15158300.
[40] W. Concepcion, Hybrid Quantum Transformer for Natural Language Processing A hybrid transformer model using a quantum kernel as the attention mechanism Master's thesis in Complex Adaptive Systems." [Online]. Available: www.chalmers.se.
[41] M. Sumithra, G.S.P. Kiruba, J. Srinivasan, M. Aishwarya, S. Sathya, and K. Geetha, Large Language Model AI Text Generation Detection based on Transformer Deep Fast Quantum Convolutional Neural Networks, in *Proceedings of the 6th International Conference on Inventive Research in Computing Applications, ICIRCA* 2025, Institute of Electrical and Electronics Engineers Inc., 2025, pp. 997–1003. doi: 10.1109/ICIRCA65293.2025.11089509.
[42] P.C. Theocharopoulos, P. Anagnostou, A. Tsoukala, S.V. Georgakopoulos, S.K. Tasoulis, and V.P. Plagianakos, Detection of Fake Generated Scientific Abstracts, in *Proceedings - IEEE 9th International Conference on Big Data Computing Service and Applications,* BigDataService 2023, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 33–39. doi: 10.1109/BigDataService58306.2023.00011.
[43] X. Yin, W. Zhang, W. Zhu, S. Liu, T. Yao, Improving sentence representations via component focusing, Appl. Sci. (Switzerland) 10 (3) (2020), https://doi.org/10.3390/app10030958.
[44] Q. Ma, C. Hao, N. Si, D. Qu, Quantum architecture search for optimizing quantum generators in quantum GAN, Mach. Learn Sci. Technol. 6 (3) (2025) 035061, https://doi.org/10.1088/2632-2153/ae056d.
[45] S. Altares-López, A. Ribeiro, J.J. García-Ripoll, Automatic design of quantum feature maps, Quantum Sci. Technol. 6 (4) (2021) 045015, https://doi.org/10.1088/2058-9565/ac1ab1.
[46] D. Lee, X. Huang, H. Hassani, and E. Dobriban, T-Cal: An Optimal Test for the Calibration of Predictive Models, 2023.